Project Design Phase-II

Data Flow Diagram & User Stories

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Data Flow Diagram — Level 0

This Level 0 flow represents the Tableau Public workflow for the fuel-price analytics project. It reflects a file-based dataset and Tableau dashboard; it does not assume a separate database or custom web application.

Entity / Process / Store	Input / Action	Output / Data Flow
External Entity: Project User / Analyst	Selects or opens the fuel-price analysis.	Questions to analyze: country, region, time, fuel prices, and tax.
Data Source: Fuel-price dataset (CSV/Excel; confirm actual format)	Provides Country, Date, Region, Income Level, Subsidy Level, crude oil, petrol, diesel, LPG, and Tax Percentage fields.	Structured fuel-price records.
Process 1: Connect & Validate Data	Connect source in Tableau; inspect data types, missing values, dates, country labels, and duplicate records.	Validated dataset for analysis.
Process 2: Analyze & Visualize	Apply suitable aggregations, filters, calculated measures, and charts.	Country comparisons, trends, regional analysis, map, and tax-price scatter plot.
Data Store: Tableau Workbook	Stores worksheets, dashboard, story, and configuration.	Reusable analysis workbook.
Output: Published Tableau Public Dashboard	Displays charts, filters, tooltips, and story points.	Interactive visual insights shared via public URL.

User Stories

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
Student / Analyst	Data Access	USN-1	As an analyst, I can open the fuel-price dataset in Tableau so that I can analyze the available fields.	The workbook connects to the source and displays the expected fields without connection errors.	High	Sprint-1
Student / Analyst	Data Quality	USN-2	As an analyst, I can check missing values, data types, dates, and country names so that comparisons are reliable.	Data issues are inspected and any corrections are documented.	High	Sprint-1
Dashboard User	Country Comparison	USN-3	As a user, I can compare average petrol and diesel prices across countries.	The charts show country labels, units, and appropriate aggregations.	High	Sprint-2
Dashboard User	Trend Analysis	USN-4	As a user, I can view fuel price trends over time.	The line chart uses Date and a clearly labelled fuel-price measure.	High	Sprint-2
Dashboard User	Interactive Filters	USN-5	As a user, I can filter the dashboard by region, date, or income level where available.	Selected filters update the relevant charts as configured.	Medium	Sprint-2
Dashboard User	Tax Analysis	USN-6	As a user, I can explore the relationship between tax percentage and petrol price.	The scatter plot shows tax and price measures, and the description avoids claiming causation.	Medium	Sprint-2
Project Reviewer	Published Report	USN-7	As a reviewer, I can open the published Tableau Public workbook to inspect the project.	The shared link opens the published workbook and the dashboard/story are visible.	High	Sprint-3